

National Neighborhood Data Archive (NaNDA): Religious, Civic, and Social Organizations by Census Tract, United States, 2003-2017



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Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains measures of the number and per capita density of select types of religious, civic, and social organizations—such as churches, mosques, synagogues, fraternal organizations, ethnic associations, and veterans’ associations—per United States census tract from 2003 through 2017.

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Data Sources

The North American Industry Classification System (NAICS) was used to identify religious, civic, and social organizations. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, Puerto Rico, and the US Virgin Islands. Other US island territories are excluded.

Methodology

This dataset highlights the concentration of churches, social clubs, fraternal societies, and other religious, civic, and social organizations per census tract in the United States. These types of organizations—along with other gathering places such as senior centers, libraries, coffee shops,

and retail stores—function as common “third places” in the United States (Finlay et al., 2018). Third places are “physical locations outside of the home (first place) or work/school (second place) that facilitate social interaction, community building, and social support” (Finlay et al., 2019). They are social infrastructure vital to civic life (Klinenberg, 2018).

In addition to their role as third places, these organizations can serve as a locale for providing health care services and health information to underserved communities (Brewer & Williams, 2019, Chaudhary et al., 2019). They also play a role in fostering a sense of civic trust and engagement (Fernandez & Alexander, 2017). Their effects on community health can also be detrimental, such as during the COVID-19 pandemic when some churches were sites of super-spreader transmission events (James et al., 2020, Woodward, 2020).

Our research team identified the following NAICS codes as representing establishments of interest:

- 8131: religious organizations, including synagogues, mosques, shrines, temples, and churches. This category excludes establishments operated by religious organizations but not focused on religious activities, such as schools, hospitals, and secondhand stores.
- 8134: civic and social organizations, including social clubs, veterans’ organizations, ethnic organizations, sororities and fraternities, and alumni associations.

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment’s census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](#). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic](#)

[Characteristics of Census Tracts, United States, 2008-2017](#). Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all organizations per tract in each year.

Usage Notes

Religious, civic, and social organizations are one of many common third places in communities. Data users interested in third places and community resources might find further useful data in these additional NaNDA datasets:

- [Arts, entertainment, and recreation establishments by census tract, United States, 2003-2017](#) (Finlay et al., 2020a), which includes gathering places such as gyms, libraries, museums, arcades, bowling alleys, and golf courses, among others.
- [Eating and drinking places by census tract, United States, 2003-2017](#) (Esposito et al., 2020a), which includes coffee shops, fast food restaurants, and bars.
- [Grocery stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020b).
- [Personal care services and laundromats by census tract, United States, 2003-2017](#) (Esposito et al., 2020b).
- [Post offices and banks by census tract, United States, 2003-2017](#) (Finlay et al., 2020c).
- [Retail establishments by census tract, United States, 2003-2017](#) (Finlay et al., 2020d), which includes stores selling furniture, clothing, sporting goods, home and garden supplies, books, and used merchandise, among other items.
- [Social services by census tract, United States, 2003-2017](#) (Finlay et al., 2020e), which includes youth and senior centers.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>.

Variable	Type	Unique	Mean	Min	Max	Label
tract_fips10	string	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	14	2009.571	2003	2017	Year
population	float	250271	4208.894	0	65528	Tract population
aland10	float	72534	48.07265	0	85425.73	Tract land area, square miles
count_8131	int	105	7.234283	0	132	# religious organizations
count_sales_8131	int	104	6.830522	0	129	# religious organizations w/ sales>0
count_emp_8131	int	68	4.166863	0	81	# religious organizations w/ 2+ employees
popden_8131	float	412703	3.358566	-1	198221.6	# religious organizations per 1000 people
popden_sales_8131	float	407708	3.181678	-1	198221.6	# religious organizations per 1000 people w/ sales>0
popden_emp_8131	float	348243	1.998233	-1	99110.82	# religious organizations per 1000 people w/ 2+ employees
aden_8131	float	339122	7.899622	0	2096.736	# religious organizations per sq mile
aden_sales_8131	float	330971	7.42635	-1	2036.829	# religious organizations per sq mile w/ sales>0
aden_emp_8131	float	303079	4.457418	0	1082.52	# religious organizations per sq mile w/ 2+ employees
count_8134	int	144	2.276429	0	248	# civic/social organizations
count_sales_8134	int	129	2.051246	0	232	# civic/social organizations w/ sales>0
count_emp_8134	int	137	2.062901	0	236	# civic/social organizations w/ 2+ employees
popden_8134	float	261957	3.035999	-1	507204.4	# civic/social organizations per 1000 people
popden_sales_8134	float	252460	2.916276	-1	507204.4	# civic/social organizations per 1000 people w/ sales>0
popden_emp_8134	float	246302	2.943494	-1	507204.4	# civic/social organizations per 1000 people w/ 2+ employees
aden_8134	float	215158	2.944876	0	1295.875	# civic/social organizations per sq mile
aden_sales_8134	float	207992	2.631545	0	1226.089	# civic/social organizations per sq mile w/ sales>0
aden_emp_8134	float	210002	2.695397	0	1262.647	# civic/social organizations per sq mile w/ 2+ employees

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